# Antlings Internship Program

# Technical Assessment – AI/ML

Drone Human Detection & Counting System

Deadline: 11:59 PM, 16 May, 2026

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## Assessment Overview

This assessment is designed to evaluate your:

- AI/ML fundamentals
- Computer vision understanding
- Problem-solving ability
- Implementation approach
- Engineering reasoning

We are more interested in your workflow, understanding, experimentation, and implementation quality. Creativity and thoughtful improvements within the scope of the task will be valued positively.

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## Task Description

Build a computer vision pipeline capable of analyzing drone/aerial images to:
• detect humans and cars
• count total humans
• visualize the final outputs

**Candidates may additionally implement object tracking as a bonus feature.**

The assessment should demonstrate your workflow from:
• dataset understanding
• preprocessing
• model training
• inference
• visualization
• evaluation

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## Dataset

A drone/aerial dataset will be provided for this assessment.

Dataset Link:
[https://www.kaggle.com/datasets/banuprasadb/visdrone-dataset?resource=download](https://www.kaggle.com/datasets/banuprasadb/visdrone-dataset?resource=download)

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## Requirements

**Task-01: Dataset Understanding & Preprocessing**

Briefly explain:
- dataset structure
- preprocessing/augmentation steps
- challenges noticed in the dataset

Include sample visualizations.

**Task-02: Model Training (Mandatory)**

Train or fine-tune an object detection model for human and car detection.

You may use:
- YOLO
- Faster R-CNN
- SSD
- RT-DETR
- or similar frameworks

Show:
- training approach
- sample predictions/results

**Task-03: Human & Car Detection with Human Counting**

Your system should:
- detect humans and cars
- display bounding boxes
- display total human count

The counting logic can be simple.

**Task-04: Optional: Object Tracking (Bonus)**

You may additionally implement tracking using:

- DeepSORT
- ByteTrack
- BoT-SORT
- OC-SORT
- or similar approaches

Tracking implementations and creative improvements will be valued positively.

**Task-05: Evaluation & Visualization**

Show:

- prediction outputs
- counting visualization
- processed images/results

Briefly discuss:

- strengths
- limitations
- challenges faced

You may optionally include metrics such as:

- precision
- recall
- mAP
- FPS

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## Deliverables

Submit:

- Public GitHub Repository (including source code, README, and outputs/results)
- Short Demo Video (3–5 minutes) showing your final implementation and results

If tracking is implemented, include tracking outputs as well.

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## Guidelines

• AI tools (ChatGPT, Claude, Copilot, Cursor, etc.) are allowed.
• However, candidates should fully understand and be able to explain their implementation during the demo/viva.

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## Evaluation Criteria

| Criteria | Weight |
|---|---|
| Dataset Understanding & Preprocessing | 20% |
| Training & Detection Pipeline | 30% |
| Counting Logic & Visualization | 20% |
| Problem Solving & Analysis | 15% |
| Code Quality & Documentation | 10% |
| Demonstration & Communication | 5% |

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## Submission Instructions

**Please submit:**
• GitHub Repository Link
• Google Drive folder link containing Demonstration Video

to: Click Here

For any task-related queries, please contact us via WhatsApp at: +880 1979-447493(Only text)

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Best of luck!
— **ANTS**